A Noninvasive Glucose Monitoring SoC Based on Single Wavelength Photoplethysmography

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***Abstract*—Conventional glucose monitoring methods for the growing numbers of diabetic patients around the world are invasive, painful, costly and, time-consuming. Complications aroused due to the abnormal blood sugar levels in diabetic patients have created the necessity for continuous noninvasive glucose monitoring. This article presents a wearable system for glucose monitoring based on a single wavelength near-infrared (NIR) Photoplethysmography (PPG) combined with machine-learning regression (MLR). The PPG readout circuit consists of a switched capacitor Transimpedance amplifier with 1 MΩ gain and a 10-Hz switched capacitor LPF. It allows a DC bias current rejection up to 20 μA with an input-referred current noise of 7.3 pA/$\sqrt{}$Hz. The proposed digital processor eliminates motion artifacts, and baseline drifts from PPG signal, extracts six distinct features and finally predicts the blood glucose level using Support Vector Regression with Fine Gaussian kernel (FGSVR) MLR. A novel piece-wise linear (PWL) approach for the exponential function is proposed to realize the FGSVR on-chip. The overall system is implemented using a 180 nm CMOS process with a chip area of 4.0 mm^2 while consuming 1.62 mW. The glucose measurements are performed for 200 subjects with R^2 of 0.937. The proposed system accurately predicts the sugar level with a mean absolute relative difference (mARD) of 7.62%.**

***Index Terms*—Glucose Monitoring, machine learning regression, near-infrared (NIR), Photoplethysmography (PPG), wearable devices.**

I. Introduction

THE RISING prevalence of diabetes and impaired glucose tolerance is evident around the world. According to the International Diabetes Federation (IDF) Diabetic Atlas [1], approximately 425 million people around the world are diabetic and 1 in 6th births is affected by hyperglycemia in pregnancy. Continuous blood glucose level (BGL) monitoring is essential for diabetic patients and pregnant women to avoid diabetes-related complications. For frequent usage, the conventional method of "finger-prick" that measures the sugar level by drawing blood samples from the finger is painful and potentially may cause various types of infections. Thus, there is an indispensable need to develop a non-invasive glucose monitoring device that measures BGL comfortably and painlessly.

Various noninvasive BGL monitoring systems have been presented in the literature based on different techniques. These techniques include impedance spectroscopy, ocular spectroscopy, raman spectroscopy, reverse iontophoresis, multi-wavelength near-infrared (NIR) spectroscopy, and other optical methods [2]–[18]. Impedance spectroscopy monitors the changes in electrical properties of the human tissues that are affected by BGL using multiple electrodes. Thus, the skin-electrode contact impedance can deteriorate the measurements [4]–[6]. The ocular spectroscopy technique utilizes a special contact lens to measure the glucose level in tears [7]. This approach has its limitation as glucose level in tears may not truly depict glucose levels in blood and contact lens are not comfortable for many diabetic patients [7]–[9]. Raman spectroscopy is another method that utilizes the laser to induce vibrations and rotations in solute molecules of a solution [10]. This theory has been used to estimate glucose concentration in human fluids. However, the instability of the laser wavelength, light intensity, long acquisition time and impractical measuring device, limit this technique for further development [10]. In reverse iontophoresis technique, low electrical current flows through the skin between an anode and a cathode placed on the skin surface. The major drawbacks of this technique are that it irritates the skin and it requires a minimum duration to obtain the measurement of BGL [11]. Furthermore, it cannot be used while the patient is sweating. NIR spectroscopy is another optical method used for noninvasive glucose monitoring [14]–[18]. It depends on capturing the transmitted or reflected light (wavelength 700–1500 nm) after penetrating through the patient skin such as fingertip or earlobe. The human body components such as muscles, skin tissues and venous blood partially absorb the light and the changes in arterial blood volume will induce periodic fluctuations in the light intensity forming the Photoplethysmography (PPG) signal [13]. Several studies have analyzed the PPG signals with different wavelengths and extracted physiological parameters of human blood to estimate blood pressure, heart rate and its variability, respiratory rate, atrial stiffness and BGL [12]–[14]. The study in [14] compared PPG signals with wavelengths in the visible and near-infrared ranges to evaluate their potential of providing reliable BGL predictions. However, most of these systems have not been optimally designed as wearable devices for measuring BGL [5]–[12].

The study in [19] used the PPG signal for identifying diabetes. PPG signals from diabetic and healthy subjects were

Manuscript received November 20, 2019; revised January 28, 2020; accepted February 25, 2020. Date of publication March 9, 2020; date of current version May 27, 2020. This work was supported by the Higher Education Commission (HEC), Pakistan, under Grant 7982/Punjab/NRPU/R&D/HEC/2017. This paper was recommended by Associate Editor Prof. J. Ohta. *(Corresponding author: Wala Saadeh.)*

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Digital Object Identifier 10.1109/TBCAS.2020.2979514

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recorded and utilized to find the parameters of the Auto Regressive Moving Average (ARMA) models. These models have then been used to classify new subjects as healthy or diabetic. Similarly, the study in [20], recorded PPG signals from healthy and diabetic subjects for classifying Diabetes Mellitus (type 2). Several features were extracted from these PPG signals and Support Vector Machines (SVM) models were developed for classification with an accuracy of 82% [20]. In [5], and [21], NIR spectroscopy was applied for estimating blood glucose. The PPG signals obtained at NIR wavelengths of 1070 nm, 950 nm, and 935 nm were processed for estimating the glucose levels, and double regression analysis was performed using an artificial neural network. In our previous work in [22], we proposed a noninvasive BGL monitoring system using PPG implemented on a printed circuit board (PCB) and Field-programmable gate array (FPGA). We recorded PPG signals of one-minute duration and processed them using various Machine learning regression models such as linear regression, SVM and Random forests [22].

Feature extraction and selection are of paramount importance for the accurate estimation of BGL [22]–[28]. These features include Kaiser Teager energy (KTE), Spectral entropy (SE), Logarithmic Energy (LogE), and Autoregressive model (AR). They determine changes in the shape of the PPG signal, spectral envelop, damping of pulses, energy profiles, and power spectrum. The statistical parameters such as mean, variance, skewness, and interquartile range are computed for these features and a feature vector is formed which is fed to the machine-learning module to estimate the BGL.

Other important aspects of designing a wearable noninvasive BGL prediction system include high accuracy measurement, patient comfort, minimum power consumption, and small size. This paper presents a noninvasive BGL monitoring system on chip (SoC) based on a single wavelength implemented in 180 nm CMOS. It extracts six selective features from the PPG signal to predict the BGL using Support Vector Regression (SVR) with Fine Gaussian kernel function. It achieves a high accuracy glucose estimation with minimum power consumption. The paper also presents a low-power and low-noise NIR PPG readout circuit based on a switched capacitor trans-impedance amplifier (TIA) to offer high gain (1 MΩ) without using a large feedback resistor.

The paper is organized as follows. Section II describes the proposed glucose monitoring system, with LED driver and PPG readout. The implementation details of PPG signal processing with feature extraction and glucose prediction are described in Section III. Section IV elaborates on the measurements and verification results. Finally, Section V concludes the paper.

II. Proposed Glucose Monitoring System

For generating a PPG signal, light is transmitted through the human tissue from NIR LED having a wavelength ($\lambda = 940$ nm). The glucose molecules in the blood vessels absorb light at this wavelength, thus, it is utilized for the glucose estimation [2]. Fig. 1 displays the working principle of the proposed noninvasive BGL estimation system. When light penetrates through the finger tissue, the glucose molecules absorb the light and

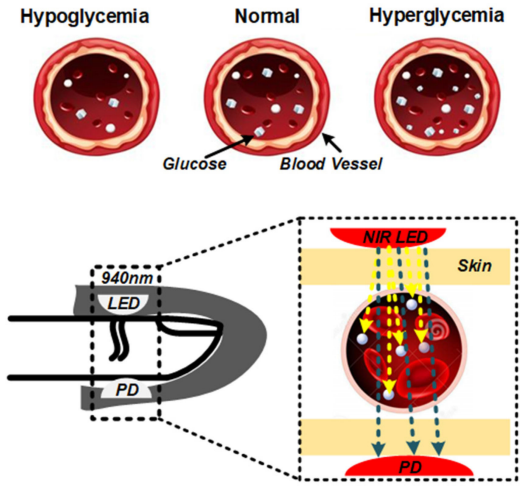


Fig. 1. Glucose in the blood vessels in hypoglycemia, normal and hyperglycemia subjects, and the working principle of the proposed BGL estimation system.

the photodiode produces a current directly proportional to the fluctuations in light intensity received by the photodiode. The block diagram of the proposed BGL estimation system is presented in Fig. 2. The LED driver provides short light pulses to reduce power consumption. The PPG readout circuit acquires the PPG signals by converting the low photodiode current to voltage through an integration (INT) mode readout utilizing a switched capacitor circuit to implement a TIA with a maximum gain of 1 MΩ.

In order to get rid of the DC component of the PPG signal, a 5-bits current Digital-to-Analog converter (DAC) at the input of the PPG readout circuit is designed to provide a DC current rejection for a maximum current of 20 μA. During the OFF-periods of the LED, the PPG readout measures the contribution of the ambient light. When the LED is turned ON, the front-end circuitry measures photodiode current, which contains large DC, small AC, and ambient light-induced components. A 5-bit code is generated in the digital feedback out of the 10-bit Analog-to-digital converter (ADC) code. The 5-bit digitized code generated in the first stage is then deducted from the current code in the DSP unit to obtain the DC component of the PPG signal, assuming that the AC component of the PPG signal is very small. Finally, this DC current is programmed into a 5-bit DAC to be connected directly into the input node for DC current removal from the photodiode current. The amplified output voltage will be then filtered using a Switched Capacitor Low pass filter (LPF) with a cut off frequency of 10 Hz to eliminate high-frequency noises. After that, the noise-free signal will be digitized using a 10-bit Successive approximation SAR-ADC [6]. Digital feedback helps to adjust the LED brightness by changing the LED ON-time and controlling the value of the DC rejection at the INT amplifier input to ensure the correct operation of the system. Next, the signal is fed to the preprocessing block, which mitigates the baseline drift and motion artifact noises. The PPG signal is divided into frames and features are extracted from the consecutive frames. Finally, the BGL value is estimated using a machine-learning regression (MLR) model.

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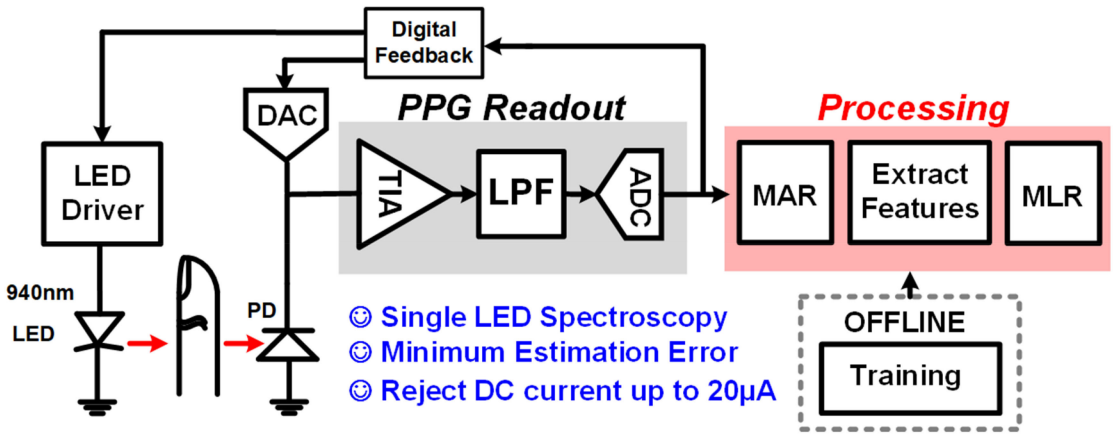


Fig. 2. The proposed noninvasive glucose estimation system's block diagram.

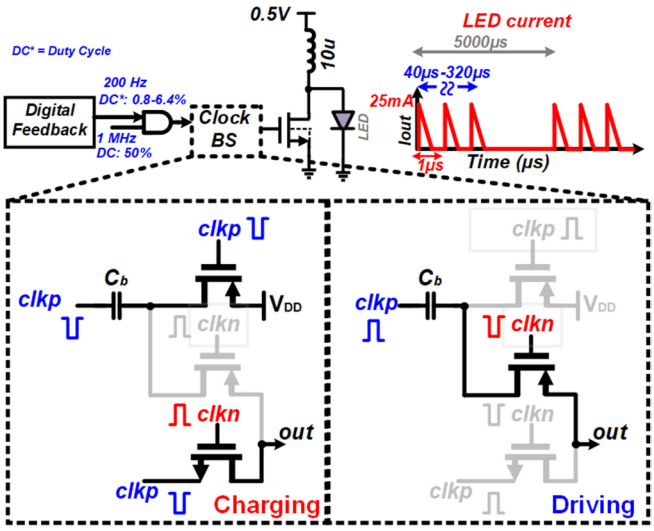


Fig. 3. LED driver with the proposed clock BS.

A. LED Driver

The NIR LED driver drives a 1.2 V NIR LED from 0.5 supply voltage using an improved form of the Boost converter as shown in Fig. 3 [29]. The NIR LED operates as a switch leading to pulsed current operating at 1 MHz. The LED brightness and power consumption are controlled by changing the ON-time of the LED through the help of digital feedback from the PPG readout circuit. The nominal ON-time of the LED is selected to be (40 μs) while operating at a pulse repletion frequency (PRF) of 200 Hz, with a variable duty cycle of 0.8%–6.4% to reduce power consumption. The LED forward current is a triangular wave with a peak current of 25 mA and an average current over 1 μs period of 7 mA. The relative radiant intensity of the LED will be similar to an LED with a constant forward current of 7 mA. Having a minimum on-resistance of the switch M1 is critical to improving the efficiency of the LED driver. Therefore, we employ a clock bootstrapping circuit (BS) to reduce the switch's on-resistance without the need for a large switch width (>1 mm). The BS circuit raises the clock pulse signal more than the supply voltage (>0.5 V). The BS operates by two non-overlapping clock phases, clk_p, and clk_n. In the charging phase as shown in Fig. 3, where clk_n is high and clk_p is low, the capacitor C_b is charged to (V_DD - V_DS). In contrast, during the driving phase, clk_n is low and clk_p is high, the series combination of the clock pulse and the voltage previously-stored on C_b will be summed together to form a larger pulse at the gate of the switch. The proposed BS achieves similar performance to the design in [30] while reducing the number of switches to three.

B. PPG Readout Circuit

Resistor-based TIAs utilize large resistors (tens of kΩ- few MΩ range) connected in the feedback of the Op-amp to generate a high impedance gain [31]–[33]. However, using large resistors will require a large die area and will raise the thermal noise of the amplifier. Hence, the proposed PPG readout circuit utilizes an INT amplifier to amplify the photodiode current without a large feedback resistor [34], [35]. The schematics of the INT amplifier is presented in Fig. 4(a). In order to operate the TIA at 0.5 V supply, the core op-amp is operated in the subthreshold region.

Careful INT amplifier design is essential to overcome undesirable effects such as charge injection and switching noise. In this implementation, the principle of correlated double sampling (CDS) is adopted to reject the undesired 1/f noise and amplifier offset [36]. The output voltage of the INT amplifier is inversely proportional to the feedback capacitor value and it depends on the total input charge that flows through during the designed clock period. The INT amplifier also contains a sample and hold (S&H) circuit at the output to save the output voltage until the next clock cycle. The INT amplifier employs three switches working at three phases of clock signal Φ1, Φ2, and Φ3. The timings diagram of these phases is shown in Fig. 4(b).

The capacitors C_f, C_out, and the clock cycle T are designed according to the desired gain of the INT amplifier. The INT amplifier works in two modes of operation, reset mode and integration mode. During the reset mode: both Φ1 and Φ2 switches are closed.

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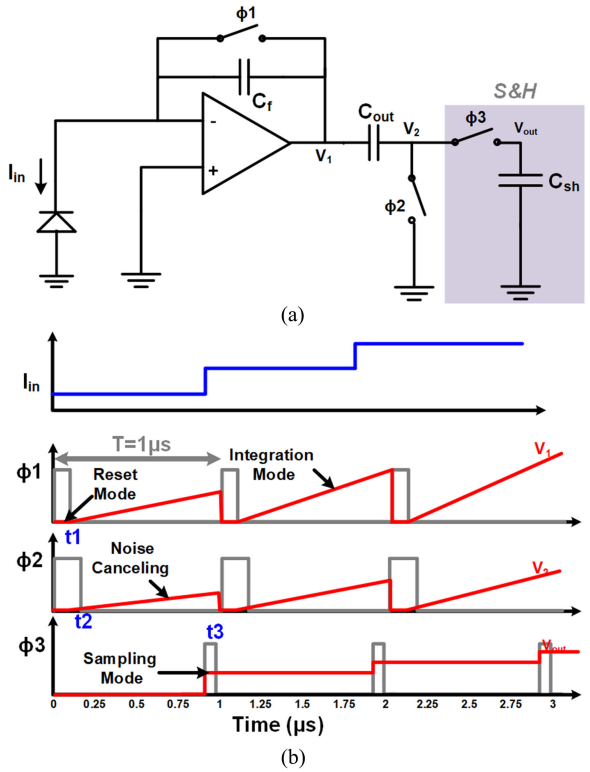


Fig. 4. (a) Schematics of INT amplifier and (b) timing diagram and operation.

Therefore, we can consider V1 as a virtual ground while the V2 is a real ground which resets the charge on both capacitors C_f and C_{out} to zero. During the integration mode, the process is distributed into three phases:

a) Initially, Φ1 is kept open while Φ2 is closed: During this phase, the input photodiode's current will be integrated on capacitor C_f, generating an output voltage V1 proportional to the input current. V2 stays at 0 since Φ2 is still connected to ground while the offset, as well as flicker noise, are stored on C_{out}.

b) In the next phase, Φ1 and Φ2 are open and the input current continues to be integrated on C_f. Since Φ2 is open now, the voltage V2 is not shorted to ground anymore; consequently, it tracks the value of V1, after detracting the offset and flicker noise stored at C_{out} according to the mechanism of CDS to cancel flicker noise and circuit offset.

c) The S&H operates when Φ3 is closed. During this phase, Φ3 is closed for a short period, the S&H circuit samples V2 and the output V_{out} is held until the subsequent time Φ3 closes.

The final output voltage of INT amplifier at each clock cycle is given by the following expression [34], [35]:

$$V_{out}(nT) = \frac{1}{C_f}\int_{t_2+(n-1)T}^{t_3+(n-1)T} I_{in}\, dt \qquad (1)$$

To provide a variable gain 100 kΩ–1 MΩ, C_f is selected to be 1pF while the clock is used to control the gain by varying the clock from 1 MHz – 10 MHz. The INT amplifier has an input-referred current noise of 7.3 pA/$\sqrt{}$Hz.

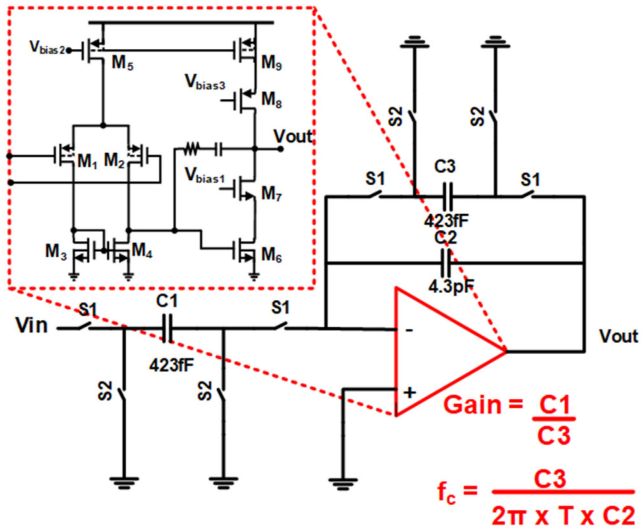


Fig. 5. Schematic of SC stray-insensitive LPF.

To realize a low cutoff frequency LPF on-chip (10 Hz), a SC stray insensitive LPF is implemented [37]. The resistor is realized using a SC circuit, which creates a large resistor without actually having it on-chip. The schematics of the SC stay-insensitive LPF is depicted in Fig. 5. The parasitic insensitive LPF has four switches operated by two non-overlapping clocks. During the first phase, S1 switches are closed, thus, the capacitor C1 charges to the Vin and the charge will be also shifted to the capacitor C2, which still has the charge stored from the previous cycle. In the second phase, when S1 switches are open while S2 switches remain closed, the capacitor C1 is now discharged since both terminals are connected to the ground. The parasitic capacitances at both the input and near to C2 will be charged and discharged to the ground, thus causing no charge transfer. Consequently, the circuit is insensitive to parasitic capacitances. By shunting the integrating capacitor with a SC-based resistance, a damped behavior is attained to deliver a finite DC gain. The ratio of C1/C3 is selected to be 1 to provide unity DC gain.

III. PPG Digital Signal Processor

A. Preprocessing of PPG Signal

The signal preprocessing of the PPG signal is essential as it can be easily corrupted with motion artifacts, baseline drifts, and other noises. This can affect the quality of PPG signals and the extracted features. Although there are many filtering techniques discussed in previous literature [38]–[40] that remove noises effectively, we aim to select a filtering technique that is computationally feasible for hardware implementation.

Filters used for noise removal in recent work [38]–[40], include digital smoothing polynomial filters (DPF), wavelet-based filters and moving average (MAV) filters. Typically, DPF and wavelet-based filters are computationally complex and more hardware costly compared to MAV filters.

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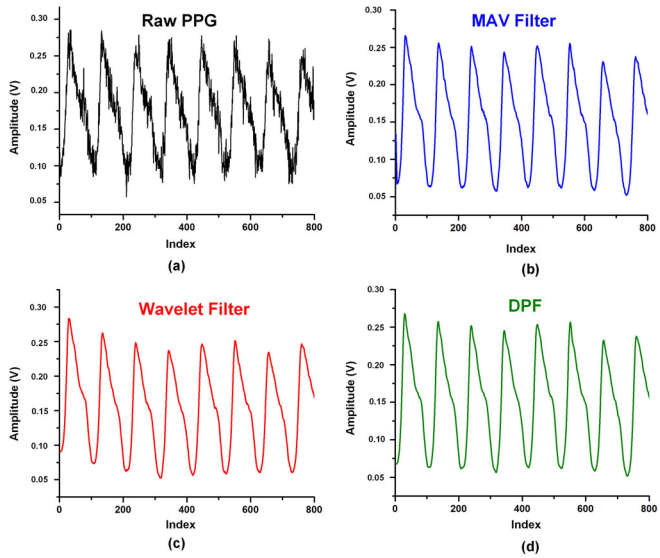


Fig. 6. (a) Raw PPG signal, and filtered PPG signal using (b) MAV filter, (c) wavelet-based filter, and (d) DFP of order 9.

To identify the optimum filter for this application, we analyzed the performance of these filters on a noisy PPG signal as shown in Fig. 6(a). We compared MAV filter with windowing length (M = 5), (Fig. 6(b)), Daubechies wavelet (db6) filter (Fig. 6(c)), and DPF of order 9 (Fig. 6(d)). All of these filters successfully rejected the noise; hence, MAV is utilized in this work for signal preprocessing. The MAV filter for M point is given by (2) [38]:

$$\mathrm{y}\left[\mathrm{n}\right] = \frac{1}{\mathrm{M}} \sum_{\mathrm{k}=0}^{\mathrm{M}-1} \mathrm{x}\left[\mathrm{n}-\mathrm{k}\right] \tag{2}$$

where $\mathrm{x}[\mathrm{n}]$ is the input signal, $\mathrm{y}[\mathrm{n}]$ is the output signal, M is the number of points on average, n is the number of samples in the input signal, and $\mathrm{k} = 0, 1, 2 \ldots . \mathrm{M} - 1$.

B. Feature Extraction

Feature extraction is a crucial part of the development of a noninvasive BGL monitoring system. The PPG signals have been studied for their use in determining heart rate and its variability (HRV), blood oxygen concentration and blood pressure [13], [21]. The physiological changes in the human body affect the shape of PPG waveform, such as changes in systolic and diastolic peak values, pulse width, peak-to-peak interval, etc. However, blood glucose concentration cannot be directly determined by calculating these changes. In order to determine BGL from the PPG signal, an indirect approach is used and distinct features are extracted. These features correlate changes in the shape of the PPG signal with the energy profile, power density, and energy spectrum of the signal. The mapping of extracted features with measured glucose levels using a machine-learning algorithm shows promising results. Different machine learning algorithms are used for training the data set and the best four are selected for comparison. The selected machine learning models are linear regression, Fine Gaussian SVR (FGSVR), SVR (quadratic) and ensemble-boosted trees.

The selection process is explained as follows: The filtered PPG signal $\mathrm{S_w}$ having a sampling rate of 128S/s of a 10s window is captured for feature extraction. This PPG signal $\mathrm{S_w}$ contains $\mathrm{L_w} = 1280$ samples that is further divided into frames $\mathrm{S_f}\ (\tau, \mathrm{n})$ where τ represents the index of the sample in the frame and $\mathrm{n} = 1, 2\ \ldots\ 10$ is the frame's number. Hence, the number of samples in each frame is $\mathrm{L_{frame}} = 128$. The frame length is a design choice due to the tradeoff between system latency, accuracy, and hardware resources. Each frame has unique values with no overlap with other frames.

Some features will be computed at the frame level while other features will be evaluated at the entire window level. The feature extraction module extracts distinct features and the input feature vector $\mathrm{X_F}$ is formed with different combinations of features. These vectors are passed to the machine-learning Matlab toolbox that trains the model and tests it on unseen data. The extracted features from PPG signals are described below:

1) Logarithmic Energy Entropy Feature: LogE is a time-domain entropy measure, computed from full-band energy spectrums [41]. LogE is computed for each frame as given in (3):

$$\mathrm{LogE} = \log\left(\sum_{\tau=1}^{\mathrm{L_{frame}}} \mathrm{S_f^2}\left(\tau, \mathrm{n}\right)\right) \tag{3}$$

Using this sequence of LogE, the statistical measures including variance (LogE_σ), mean (LogE_μ) and interquartile range $(\mathrm{LogE_{iqr}})$ are determined.

2) Kaiser-Teager Energy Feature: KTE operators are defined for both continuous and discrete domains. These operators are commonly utilized to determine the instantaneous energy profiles of signals [42]. Energy functions of complicated functions can be expressed as a product of simpler functions using KTE operators. One of the useful properties of KTE operators in the signal analysis is to indicate whether a signal is noisy or clean. The high mean value of KTE indicates a clean signal while low mean value indicates a noisy signal [23]. This property gives insight into the signal quality and can help in deciding whether to keep or discard a signal.

The KTE operator for a discrete signal $\mathrm{x}[\mathrm{n}]$ is given by

$$\mathrm{KTE}\left[\mathrm{n}\right] = \mathrm{x}^2\left[\mathrm{n}\right] - \mathrm{x}\left[\mathrm{n}-1\right]\mathrm{x}\left[\mathrm{n}+1\right] \tag{4}$$

In the discrete domain, the energy operator requires only three adjacent samples of the signal hence providing excellent time resolution and lower computational complexity [42]. KTE features for the proposed system are computed both at the window and frame level. At window level, KTE sequence is computed as

$$\mathrm{KTE_w}\left(\mathrm{k}\right) = \mathrm{S_w^2}\left(\mathrm{k}\right) - \mathrm{S_w}\left(\mathrm{k}-1\right) \times \mathrm{S_w}\left(\mathrm{k}+1\right) \tag{5}$$

for $\mathrm{k} = 2, 3\ \ldots\ \mathrm{L_w} - 1$.

At the frame level, the sequence is given by

$$\mathrm{KTE}\left(\tau, \mathrm{n}\right) = \mathrm{S_f^2}\left(\tau, \mathrm{n}\right) - \mathrm{S_f}\left(\tau+1, \mathrm{n}\right) \times \mathrm{S_f}\left(\tau-1, \mathrm{n}\right) \tag{6}$$

for $\tau = 2, 3\ \ldots\ \mathrm{L_{frame}} - 1$ and $\mathrm{n} = 1, 2 \ldots . 10$. From $\mathrm{KTE}(\tau, \mathrm{n})$, we find the mean (KTE_μ), variance (KTE_σ), interquartile range $(\mathrm{KTE_{IQR}})$ and skewness $(\mathrm{KTE_{skew}})$ for each frame.

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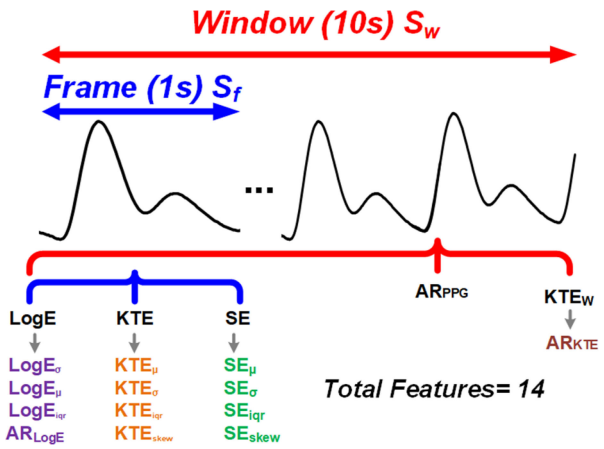


Fig. 7. PPG signal showing frame and window levels, and their corresponding features.

Fig. 7 shows the PPG signal describing the frame and window levels and their features.

3) Spectral Entropy Feature: SE is a tool for estimating the spectral power distribution of a signal. SE-based measurements indicate whether a signal is noisy or if some information is lost due to a noisy environment. SE also determines the harmonic components, spectral shape, and damping of pulses. The concept of using SE values to determine speech signals has been used in previous literature [43]–[45]. This concept can be extended to determine a range of SE values that correspond to particular values of glucose level in blood.

SE is computed in a number of steps. Initially, Fast Fourier Transform (FFT) is computed on each frame $\mathrm{S_f}\,(\tau, \mathrm{n})$ of length $\mathrm{L_{frame}} = 128$. From the time-domain, frames are converted into the frequency domain.

$$\mathrm{X_n}\,[\mathrm{k}] = \mathrm{FFT}(\mathrm{S_f}\,(\tau, \mathrm{n})) \tag{7}$$

The vector $\mathrm{X_n}[\mathrm{k}]$ is the Fourier transform of signal $\mathrm{S_f}\,(\tau, \mathrm{n})$ of length $\mathrm{k} = 128$. Secondly, the normalization of the squared absolute value of $\mathrm{X_n}\,[\mathrm{k}]$ is performed and vector $\mathrm{P_X^n}[\mathrm{k}]$ is formed.

$$\mathrm{P_X^n}\,[\mathrm{k}] = \frac{|\mathrm{X_n}\,[\mathrm{k}]|^2}{\sum_{\mathrm{j=1}}^{\mathrm{L_{frame}}} |\mathrm{X_n}\,[\mathrm{j}]|^2}, \quad \mathrm{k} = 1, 2 \ldots \mathrm{L_{frame}} \tag{8}$$

The final step computes the SE from the normalized term $\mathrm{P_X^n}[\mathrm{k}]$

$$\mathrm{SE} = \sum_{\mathrm{k=1}}^{\mathrm{L_{frame}}} \mathrm{P_X^n}\,[\mathrm{k}]\ \mathrm{Log}\ (\mathrm{P_X^n}\,[\mathrm{k}]) \tag{9}$$

The features extracted from SE are the mean (SE_μ), variance (SE_σ), interquartile range $(\mathrm{SE_{iqr}})$ and skewness $(\mathrm{SE_{skew}})$.

4) Autoregressive Model of PPG Signal: The final features are based on the AR model coefficients. The AR coefficients are widely used for respiratory rate extraction from PPG signals [46]. For BGL estimation, we compute these features at window level directly and for the previously computed $\mathrm{KTE_w}$ (window level) and LogE (frames level) features. Henceforward, forming $\mathrm{AR_{PPG}}$, $\mathrm{AR_{KTE}}$, and $\mathrm{AR_{LogE}}$ vectors. The signals $\mathrm{S_w}$, $\mathrm{KTE_w}$, and LogE are modeled using 3 different AR models given by (10), (11) and (12), respectively [46]:

$$\mathrm{S_w}\,(\mathrm{n}) = -\sum_{\mathrm{i=1}}^{\mathrm{p}} \mathrm{AR_{PPGi}S_w}\ (\mathrm{n-i}) + \mathrm{e}\,(\mathrm{n}) \tag{10}$$

$$\mathrm{KTE_w}\,(\mathrm{n}) = -\sum_{\mathrm{i=1}}^{\mathrm{p}} \mathrm{AR_{KTEi}KTE_w}\,(\mathrm{n-i}) + \mathrm{e}\,(\mathrm{n}) \tag{11}$$

$$\mathrm{LogE}\,(\mathrm{n}) = -\sum_{\mathrm{i=1}}^{\mathrm{p}} \mathrm{AR_{LogEi}LogE}\,(\mathrm{n-i}) + \mathrm{e}\,(\mathrm{n}) \tag{12}$$

Where p is the order of the AR model, $\mathrm{AR_{PPGi}}, \mathrm{AR_{KTEi}}$, and $\mathrm{AR_{LogE_i}}$ are the AR models' coefficients and $\mathrm{e(n)}$ is the prediction error in each model. The order of the coefficients is selected such that the AR power spectrum appropriately maps the periodogram of the PPG signal. In this work, AR models of order 3 are chosen based on our simulations to achieve a good tradeoff between accuracy and efficient hardware implementation.

Input vector of all fourteen distinct features is given by:

$$\mathrm{X_{F14}} = [\mathrm{AR_{PPG}}, \mathrm{KTE}_\mu, \mathrm{KTE}_\sigma\ , \mathrm{KTE_{iqr}}, \mathrm{KTE_{skew}}, \mathrm{AR_{KTE}}, \mathrm{SE}_\mu\ \mathrm{SE}_\sigma, \mathrm{SE_{iqr}}, \mathrm{SE_{skew}}, \mathrm{LogE}_\mu, \mathrm{LogE}_\sigma, \mathrm{LogE_{iqr}}, \mathrm{AR_{LogE}}]^{\mathrm{T}} \tag{13}$$

The feature vector $\mathrm{X_{F14}}$ is of length 20, since each AR feature has 3 coefficients but we are just considering them as a single feature for ease of understanding.

C. Features Selection and Machine Learning Regression

To identify the minimum feature set and the optimal machine learning regression for glucose perdition, the Wrapper method is applied on measured PPG signals and glucose readings for 200 subjects (males (112) and females (88): age (18 to 71 years) among them 11 diabetic patients [46]. The measurement procedure was approved by the Institutional Review Board (IRB) LUMS-IRB/20180409. The PPG signals are acquired using the prototype described in [22] and [48]. Out of the total 200 subjects' data, we randomly selected 60% of data for the training and the remaining 40% for validation. This process was repeated 5 times, and the model that performed the best among those has been implemented on-chip.

The accuracy of the proposed BGL monitoring system is obtained using two different analysis: 1) the mean absolute relative difference (mARD) and 2) the root mean square error (RMSE). The mARD is calculated as shown in (14)

$$\mathrm{mARD} = \frac{1}{\mathrm{n}} \sum_{\mathrm{i=1}}^{\mathrm{n}} \frac{|\mathrm{BGL_{Est.}} - \mathrm{BGL_{Mea.}}|}{\mathrm{BGL_{Mea.}}} \times 100\% \tag{14}$$

Where n is the number of test samples, $\mathrm{BGL_{Est.}}$ is the predicted blood glucose level, and $\mathrm{BGL_{Mea.}}$ is the reference BGL, measured using finger-stick off-the-shelf Accu-Chek instant S (Roche Pakistan Limited device). We evaluated all the possible combinations of the aforementioned feature vectors lengths

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TABLE I
WRAPPER METHOD FOR MINIMUM FEATURES SET FOR GLUCOSE ESTIMATION

		SVR- Fine Gaussian		SVR- Quadratic		Linear Regression		En. Boosted Trees	
Number of Features	Features selected	mARD	RMSE	mARD	RMSE	mARD	RMSE	mARD	RMSE
14	AR_{PPG}, KTE_{σ}, KTE_{μ} , KTE_{iqr}, KTE_{skew}, AR_{KTE} , $LogE_{\sigma}$, $LogE_{\mu}$,$LogE_{iqr}$, AR_{LogE}, SE_{σ}, SE_{μ}, SE_{iqr}, SE_{skew}	8.36	11.29	18.27	25.21	22.57	33.85	18.27	25.21
11	KTE_{σ}, KTE_{μ} , KTE_{iqr}, KTE_{skew}, $LogE_{\sigma}$, $LogE_{\mu}$,$LogE_{iqr}$, SE_{σ}, SE_{μ}, SE_{iqr}, SE_{skew}	10.16	12.31	15.01	46.00	14.66	26.00	15.24	21.62
10	KTE_{σ}, KTE_{μ} , KTE_{iqr}, KTE_{skew}, $LogE_{\sigma}$, $LogE_{iqr}$, SE_{σ}, SE_{μ}, SE_{iqr}, SE_{skew}	13.66	21.93	22.09	44.41	16.19	29.94	16.18	23.00
8	KTE_{σ}, KTE_{μ} , KTE_{iqr}, KTE_{skew}, SE_{σ}, SE_{μ}, SE_{iqr}, SE_{skew}	12.17	21.6	22.05	50	19.19	25.86	16.05	23.21
6	KTE_{σ}, KTE_{μ} , $LogE_{\sigma}$, $LogE_{\mu}$, SE_{σ}, SE_{μ}	7.62	11.20	21.10	42.90	13.22	23.35	9.67	13.00

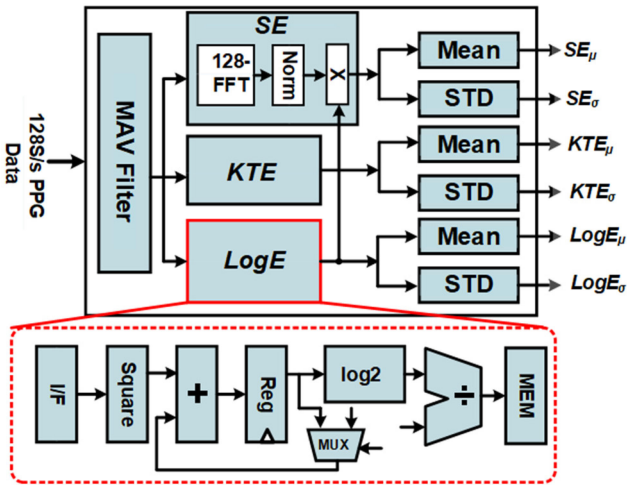


Fig. 8. Block diagram of features extraction engine.

according to the Wrapper method [49]. Since it is impossible to include all the possible combinations, in Table I we summarize the best combinations for four different machine-learning techniques. It can be noticed that the minimum value of mARD is achieved using the FGSVR algorithm with six and fourteen features, while ensemble boosted trees shows the minimum value of mARD with six features. These results imply that six features are sufficient to estimate the glucose level in blood. Hence, the FGSVR is adopted with 6-features feature-vector and the implementation details are explained in the following section.

D. Glucose Prediction Processor Implementation

Fig. 8 displays the block diagram of the proposed feature extraction (FE) engine. The FE engine accepts the digitized PPG data from the SAR-ADC at a sampling rate of 128S/s. The MAV filter removes the baseline [50] and noise of the digitized PPG signal.

Once the signal is clean, the data is processed to evaluate the six features, i.e., mean and standard deviation of SE, KTE, and LogE (KTE_{σ}, KTE_{μ}, $LogE_{\mu}$, $LogE_{\sigma}$, SE_{μ}, SE_{σ}). To ensure hardware-efficient implementation, area and power optimization techniques are applied during the synthesis and placement stages. Moreover, LogE implementation is shown in Fig. 8, which utilized area-optimized IP core of log-2 to evaluate log-10 implementation required for the LogE evaluation. The proposed technique saves area resources by 27% compared to the conventional lookup table implementation of log-10 [51].

The FGSVR for predicted $BGL_{Est.}$ is defined as (15) [52]:

$$BGL_{Est.} = \sum_{i=1}^{N} \alpha \times e^{\left(-\left(X_i - X_c{}^2/(2\sigma^2)\right)\right)} + \text{Bias} \qquad (15)$$

Where N is the number of support vectors, σ controls the variance of Gaussian basis function (GBF), and "α vector" is the weighting vector. X_i and X_c are the features' vector and support vector, respectively. The values of σ, X_c, and Bias are decided during the training phase.

The hardware implementation of the FGSVR is a direct implementation of the equation (15). However, it comes with the major hardware bottleneck of the exponential function which is implemented in the presented system with the help of the proposed floating-point piece-wise linear (FP-PWL) approach that is modified form of [53] and achieves efficient implementation compares to the [54], and [55]. The proposed approach separates the mantissa and exponent part of incoming data (18 bits). The exponential of the 18-bit floating-point input data is evaluated by separating the mantissa (m) and exponent (exp) parts as shown in Fig. 9, where 'm' is a 10-bit mantissa and 'e' is a 7-bit exponent. The exponent of the 10-bit precise mantissa ($0 \leq m < 1$) is estimated using the 10 PWL regions to ensure high precision and maximum accuracy with a minimal maximum error of 2.1%. The evaluated mantissa exponent is multiplied by "n" times where "n" is the normalized exp value for the input

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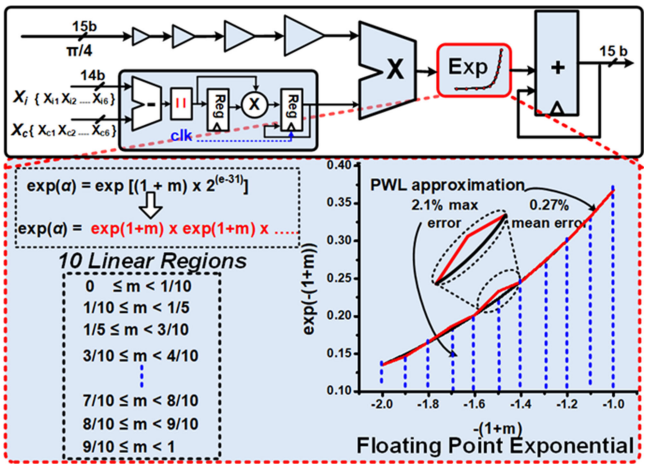


Fig. 9. Implementation of FGSVR MLR model.

data. The main novelty in the proposed implementation is to represent the exponential function as follows:

$$e^{\alpha} = e^{\left((1+m)\times 2^{e-31}\right)} \tag{16}$$

Where α is the input number and we aim to find its exponential function by expanding it into the IEEE-754 floating-point format (Right-hand side of (16)). Then we find another exponential function (through normalization) of a smaller value that lies between 1 and 2 using the PWL approach. We decide the normalization factor by finding the highest power-of-2 which guarantees a division output in the region between 1 and 2. The normalization factor $\mathbf{n}$ is decided by the normalized exponent value for the input data.

$$(1+m) = 1 \le \frac{\alpha}{2^{N}} < 2 \tag{17}$$

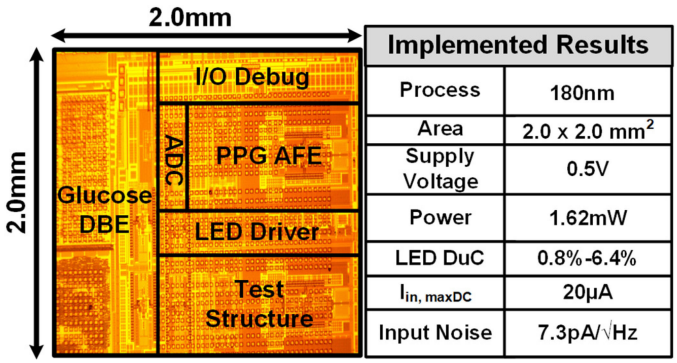
Where $\mathbf{n} = 2^{N}$. After that, we can find the original e^{α} by n-times self-multiplication of $e^{(1+m)}$ as given by (18):

$$e^{\alpha} = \left(e^{(1+m)}\right)^{n} \tag{18}$$

Where the number of times of the self-multiplication of $e^{(1+m)}$ to compute e^{α} depends upon the normalization factor n (i.e., 2^{N}). We implemented this process using adders only. The proposed approach helps to reduce the required regions of the PWL method significantly. We can find any exponential function by a simple PWL and self-multiplications implemented using the addition operation.

IV. Measurement Results and Discussion

The complete glucose monitoring system is realized in 180 nm CMOS, with an active area of 4.0 mm^2 as shown in Fig. 10. The BGL monitoring system is powered from 0.5 V supply consuming 1.62 mW, which is 90% less power compared to [5]. The PPG readout provides an overall measured signal-to-noise ratio (SNR) of 60.3 dB comparable to design in [56] at 44% reduced LED power. The PPG SoC was designed to meet the SNR requirements of >28.7 dB for mARD of 7.62%. The



Implemented Results	
Process	180nm
Area	2.0 x 2.0 mm^2
Supply Voltage	0.5V
Power	1.62mW
LED DuC	0.8%-6.4%
$I_{in,\ maxDC}$	20μA
Input Noise	7.3pA/√Hz

Fig. 10. Die photo of glucose prediction system and performance summary.

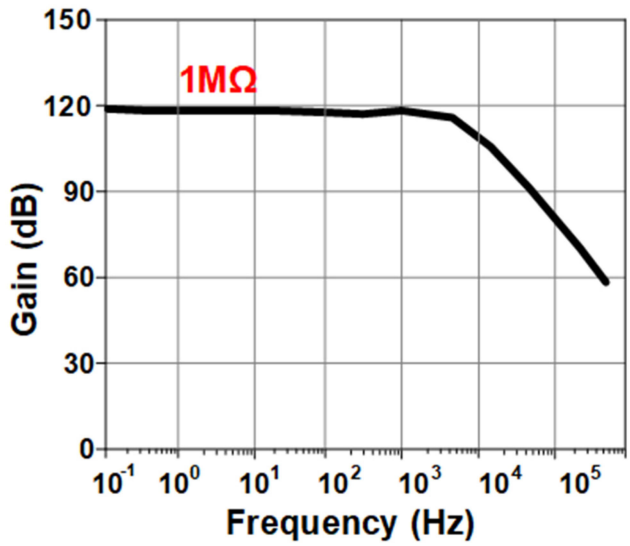


Fig. 11. Measured maximum gain and bandwidth of INT amplifier.

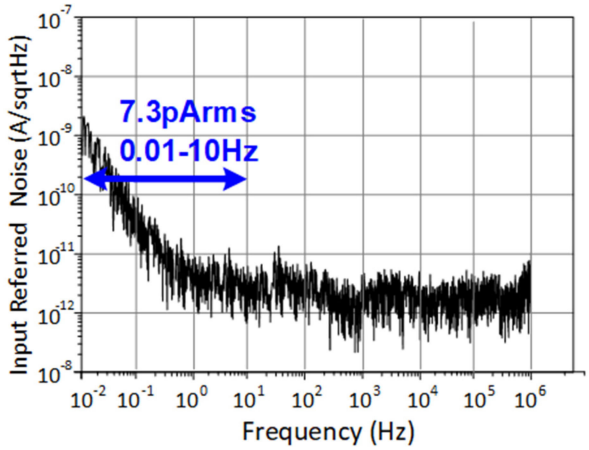


Fig. 12. The measured input referred noise of INT amplifier.

glucose estimation processor consumes 450 μW. The measured Gain-Bandwidth (GBW) of the INT amplifier is shown in Fig. 11, providing a maximum gain of 1 MΩ and bandwidth of 25 kHz while consuming 6.2 μW. The measured input-referred current noise of the INT amplifier is shown in Fig. 12, offering an integrated input-referred current noise of 7.3 pA/√Hz for

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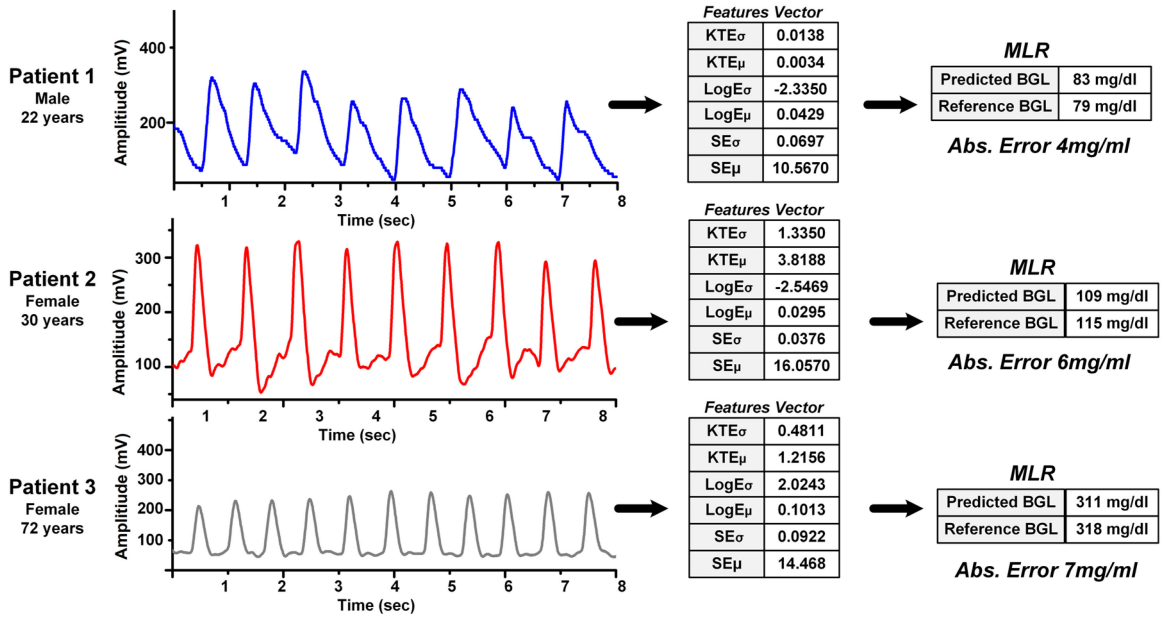


Fig. 13. Measured BGL estimation process for 3 different subjects having reference BGL of 79, 115 and 318 mg/ml, respectively.

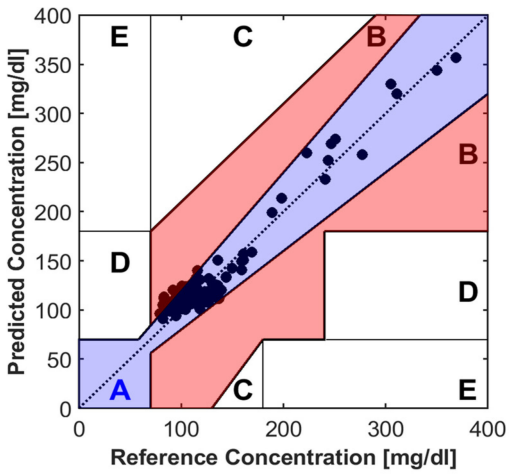


Fig. 14. The Clarke error grid analysis of estimated and reference BGL.

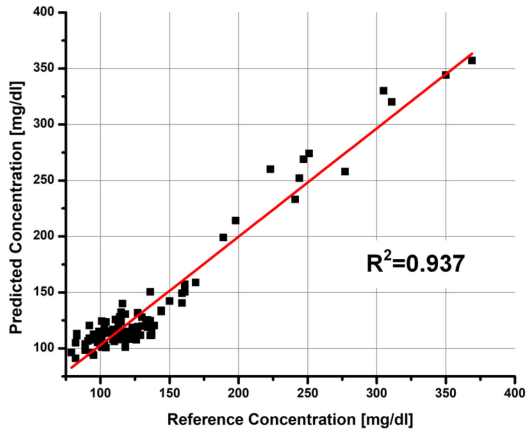


Fig. 15. Predicted versus reference glucose concentration for 200 subjects.

a bandwidth of 0.01–10 Hz, which decreases the input-referred noise by 89% compared to design in [29] at the cost of an increase in the power consumption. Additionally, with the help of DAC, it allows a maximum DC current of 20 μA versus 4 μA in [29].

The operation of the proposed system is presented for 3 different subjects in Fig. 13. The measured raw PPG signals are utilized to extract feature vectors, which are fed to MLR for BGL estimation. The predicted BGL values are very close to reference values. To further prove the accuracy of the proposed system for glucose prediction, the Clarke error grid (CGE) analysis is carried out on 200 subjects as displayed in Fig. 14. The subjects are 110M/80F with age (20–45 years). The PPG signals are measured using the proposed PPG readout circuit and they are different from the 200 subjects' recordings that were used for the training process described in Section VI. During the measurement process, the subjects are advised to sit still for acquiring the PPG signal to minimize the motion artifacts.

Then their glucose levels at that time were also measured using a commercial glucometer (Accu-Chek instant S (Roche Pakistan Limited device)). Firstly, we used Alcohol pads to sanitize the finger, and then we used a new needle for each subject to prick the skin and draw blood drops. The measured glucose values were recorded and then compared with the predicted values from our proposed glucose estimation processor. The CGE analysis provides a standard method to compare the performance of blood glucose estimation systems [5]. It has 5 main regions: A (for values limited to 20% of the reference value), B (more than 20% and doesn't cause wrong treatment), while C, D, and E regions are error regions. The proposed system shows more than 95% of predicted glucose values are in region A, which is regarded as clinically acceptable zone and with measured mARD of 7.62% and RMSE of 11.20. The remaining 5% lies in region B. The

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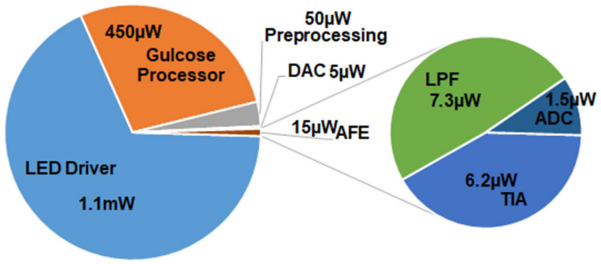


Fig. 16. The distrubtion of the power consumption of the glucose prediction system.

TABLE II
COMPARISON WITH THE STATE-OF-THE-ART WORK FOR NONINVASIVE GLUCOSE ESTIMATION SYSTEMS

	P. Pai EMBC'15 [26]	Z. Geng, Nature'17 [27]	Y. Zhang, ITOEC'17 [24]	K. Song, JSSC'15 [5]	A. Hina, ISCAS'19 [22]	This Work
Technology	Multi NIR	Multi NIR and other sensors	Multi NIR	Multi-NIR and Impedance	NIR with GPR exponential	NIR with SVR Fine
mARD	9.61%	14.4%	-	8.3%	8.97%	7.62%
RMSE	-	-	1.58	-	12.49	11.20
Number of features	8	-	22	4	10	6
Number of patients	30	9	18	10	200	200
Hardware implementation	x	x	x	O	FPGA	O
Power Consumption	-	-	-	38mW	-	1.62mW

TABLE III
COMPARISON WITH THE STATE-OF-THE-ART WORK FOR NIR PPG READOUTS

	M. Konijnenburg JSSC'16 [57]	P. Schönle JSSC'18 [58]	A. Sharma JSSC'17 [59]	J. Xu TBioCAS'18 [60]	This Work
Topology	Resistive	Resistive/ Capacitive	Resistive/ Capacitive	Resistive	Capacitive
Technology	180nm	130nm	130nm	180nm	180nm
Supply (V)	1.2	1.2	1.5	1.2	0.5
AFE Power (μW)	135	340	50	-	15
DC Cancellation (μA)	10	10	12	40	20
Dynamic Range (dB)	89	112	97	87	106
Input Noise	$15pA_{rms}$	-	$5pA_{rms}$	$0.9nA_{rms}$	$7.3pA_{rms}$

comparison of predicted results with the reference values is shown in Fig. 15, that clearly indicate an accurate prediction of the glucose values with the Coefficient of determination (R^2) of 0.937.

The distribution of power consumption is shown in Fig. 16, where the LED driver consumes most of the system's power. Table II compares the proposed system and the state-of-the-art work for non-invasive glucose monitoring. The proposed system offers less mARD compared to multi-wavelength NIR spectroscopy and other spectroscopies systems with minimum power consumption at the cost of complex MLR implementation. Table III compares the proposed NIR PPG readout with the state-of-the-art NIRs PPG readouts. The proposed readout utilizes a capacitive approach with minimum power consumption, low integrated input-referred current noise and large dynamic range.

V. CONCLUSION

A wearable system for noninvasive glucose measurement using a single wavelength NIR (940 nm) with FGSVR MLR is presented. The proposed PPG readout circuit utilizes an INT amplifier with a maximum 1 MΩ gain and a switched capacitor 10 Hz-LPF. It rejects a DC bias current up to 20 μA with a low input-referred current noise of 7.3 pA/$\sqrt{\text{Hz}}$. The proposed glucose estimation processor eliminates motion artifacts and baseline drifts. It extracts 6 features from the PPG signal to estimate the blood sugar utilizing FGSVR MLR. The feature extraction engine and the FGSVR are carefully designed and realized on hardware to achieve high accuracy with minimum power and area usage. A novel FP-PWL approach for the exponential function is proposed to realize the FGSVR on-chip. The overall system is realized using a 180 nm CMOS process with a chip area of 4.0 mm^2 while consuming 1.62 mW. The validation process is done using 200 subjects with R^2 of 0.937 and mARD of 7.62%.

V. DATA AVAILABILITY

The authors are willing to share the recorded PPG data based on reasonable requests to the corresponding author.

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